



Loan Default Prediction

CLASS PRESENTATION · MEDIUM DECK · DARK
BOTANICAL THEME

58,645

Observations

20

Core variables

85.8% /

14.2%

Class split

Loan Default Prediction

CLASS PRESENTATION

58,645

Observations

20

Core variables

Agenda

1. Business problem

Why lenders need better default screening and automated approval support.

2. Dataset and target

Applicant, loan, and credit-history features with a binary default outcome.

3. Preparation steps

Cleaning, encoding, and feature engineering to improve model readiness.

4. Insights and next steps

Class imbalance, model training, and what could strengthen the analysis.

Why this matters

- Loan approval is framed as a binary classification task.
- Financial institutions want to reduce default risk while still approving viable borrowers.
- The project aims to support faster, more data-driven lending decisions.
- Because defaults are the minority class, raw accuracy can be misleading.

14.2%

Observed default share in the project summary

Dataset snapshot

58,645

Training rows

20

Variables

0 / 1

Target (Default)

Personal data

Age, home ownership, and employment length.

Loan details

Loan amount, interest rate, intent, loan-to-income ratio.

Credit history

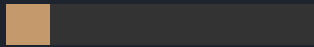
Previous default flag and credit history length.

Class imbalance drives evaluation

Non-default: 85.8%



Default: 14.2%



Why it matters

- Majority-class predictions can look strong even when defaults are missed.
- Minority detection is crucial because defaulters create the lending loss.
- Threshold tuning and recall/precision trade-offs are essential.

Feature engineering

Engineered features

Created loan_percent_income ratio; applied log transforms to income/amount.

Encoding

Loan grade ordinal-mapped (A-G); prior default-on-file binary; home ownership/intent one-hot encoded.

Modeling workflow

1. Load train, test, and sample submission files.
2. Encode and engineer features.
3. Split features and target.
4. Align train/test columns.
5. Fit the model (Logistic Regression, XGBoost, Random Forest) and generate predictions.

Visual analysis

Target count plot

Inspects the approval/default split.

Income histogram

Shows distribution of applicant income.

Correlation heatmap

Assesses relationships among numerical features.

Key takeaways

Model Strategy

Imbalanced classes require focus on F1-score and ROC-AUC rather than raw accuracy.

Future Work

Incorporate confusion matrices for decision transparency and perform threshold tuning.

Thank You

Questions on data, modeling, or evaluation?